

Mouse auditory experiment

A first plan

October 8, 2026

We want to play short tones to a mouse and record its brain responses. SONIC tested awake sheep; our mouse state and tone settings remain open. (Methods 6.6, p.13)

1. Choose the sounds

- Choose the frequency, duration, loudness, and gap between tones.
- **SONIC:** initially explored 1.4–32 kHz; benchmark tones were 10 ms sine waves with 2 ms fades at each end. (Methods 6.1–6.2, p.11)
- **SONIC:** random frequencies, balanced within each block, with no silence between tones. (Methods 6.1, p.11)
- Save the sequence before playback, as SONIC did. (Methods 6.2, p.11)

2. Set up the speaker

- Plan as though we need to assemble the whole sound setup.
- **SONIC:** NI USB-6361 output at 192 kHz, TDT SA1 amplifier, and MF1 free-field speaker; about 80 dB SPL at sheep-ear distance. (Methods 6.2, p.11)
- **Our plan:** compare the equipment on the next page, keep speaker placement fixed, and measure each tone with a calibrated microphone at the mouse-ear position.

3. Keep everything on one timeline

- Schedule each tone with a timing pulse and record it alongside the neural signals.
- **SONIC:** recorded electrical sound output and trial-start markers; shared clock references aligned these with Connexus brain recordings. (Methods 6.2, p.11)
- **Our additions:** synchronize a microphone recording to check actual sound arrival. Align video frame times too if we add behavior video.

Reference: Perkins et al. (2025), SONIC: A Benchmarking Paradigm for Brain-Computer Interfaces, bioRxiv v1. Page numbers above refer to the printed manuscript. Mouse choices and additions are proposals.

Equipment we need

Mouse setup: building from scratch

Xiaorong forwarded Tinghan's equipment list. We will use it to compare options; access to those items still needs confirming.

1. Play the tones

- **Candidate path:** computer → timed sound output (DAC/DAQ) → ED1 driver → ES1 speaker.
- **ES1:** plays sound through the air. Its intended range starts at 4 kHz; SONIC's lowest tones fall below that range. (TDT speaker specs)
- **ED1:** also needs zBus power, such as ZB1PS, and the matching speaker cable. It does not generate the tones. (TDT driver specs)
- **Other choices:** EC1 delivers sound through tubing. Peerless XT25TG30-04 and Scan-Speak R2004/602200 need a conventional amplifier; check their usable frequency range before choosing.

2. Measure the sound

- **Avisoft option:** the complete UltraSoundGate 116Hm kit includes a CM16/CMPA microphone and preamplifier. The 116Hm alone is the recording interface. (Avisoft kit details)
- **B&K option:** the 4939 microphone needs a compatible preamplifier, polarization supply, and recorder. (4939 datasheet)
- Either path needs calibration records and a sound-level reference. Check each tone at the planned ear position.

3. Record the timing and brain signals

- Choose the neural recorder and headstage, then match its event inputs to the tone-output system.
- Record tone markers in both sound and neural data. Check actual sound arrival and clock drift; a shared marker alone does not make two clocks identical.
- Include cables, power, fixed mounts, and data storage. Add a camera only if behavior is needed; align its frame times too.
- A soundproof box is not required by current team direction. Record background sound and the setup geometry.

Equipment candidates are from the forwarded BCM list; linked specifications were checked October 8, 2026. Mouse state, frequency range, and final hardware remain open.